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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 11	
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Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Docker Compose for Multi-Container App

Implementation & Theory	Project directory + base files — Created my-multi-app and carried over the Flask app, requirements, and Dockerfile from the earlier volumes assignment as a starting point.
	
	`from flask import Flask`
	`import redis`
	`import os`
	
	`app = Flask(__name__)`
	`r = redis.Redis(host='redis', port=6379, decode_responses=True)`
	
	`@app.route('/')`
	`def hello_world():`
	`  count = r.incr('hits')`
	`  return f'Hello, Docker! This page has been viewed {count} times.'`
	
	`if __name__ == '__main__':`
	`  app.run(debug=True, host='0.0.0.0', port=5000)`
	
	app.py updated for Redis — Modified the Flask app to connect to a Redis service (hostname redis, the Compose service name) and increment a hits counter on every request, returning the current count.

```
Flask==2.1.2
Werkzeug==2.1.2
redis==4.5.4
```

requirements.txt updated — Added redis==4.5.4 as a dependency alongside the existing Flask/Werkzeug pins.

```yaml
version: '3.8'
Run All Services
services:
  Run Service
  web:
    build: .
    ports:
      - "8080:5000"
    depends_on:
      - redis
  Run Service
  redis:
    image: redis:latest
    ports:
      - "6379:6379"
    volumes:
      - redis-data:/data

volumes:
  redis-data:
```

docker-compose.yml created — Defined two services: web (built from the local Dockerfile, mapped to host port 8080) and redis (official Redis image, with a named volume redis-data for persistence). depends_on ensures Redis starts before the web service.

```
WARN[0000] /Users/raghavdhoot/devops-a11/my-multi-app/docker-compose.yml: the attribute `vers
ion` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 11/11
 ✔ Image redis:latest             Pulled                                                11.1s
 ✔ Network my-multi-app_default   Created                                                0.0s
 ✔ Container my-multi-app-redis-1 Created                                                0.1s
 ✔ Container my-multi-app-web-1   Created                                                0.0s
Attaching to redis-1, web-1
redis-1  | Starting Redis Server
redis-1  | 1:C 12 Oct 2026 11:55:29.979 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
redis-1  | 1:C 12 Oct 2026 11:55:29.979 * Redis version=8.10.2, bits=64, commit=00000000, mod
ified=1, pid=1, just started
redis-1  | 1:C 12 Oct 2026 11:55:29.979 * Configuration loaded
redis-1  | 1:M 12 Oct 2026 11:55:29.980 * monotonic clock: ARM CNTVCT @ 24 ticks/us
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * Running mode=standalone, port=6379.
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf> RedisBloom version 8.10.1 (Git=unknown)
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf> Registering configuration options: [
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { bf-error-rate       :      0.01 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { bf-initial-size     :       100 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { bf-expansion-factor :         2 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { cf-bucket-size      :         2 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { cf-initial-size     :      1024 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { cf-max-iterations   :        20 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { cf-expansion-factor :         1 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf>  { cf-max-expansions   :        32 }
redis-1  | 1:M 12 Oct 2026 11:55:29.981 * <bf> ]
redis-1  | 1:M 12 Oct 2026 11:55:29.982 * Module 'bf' loaded from /usr/local/lib/redis/module
s//redisbloom.so
redis-1  | 1:M 12 Oct 2026 11:55:29.988 * <search> search-workers default: 8 (min of MAX_WORK
ER_THREADS=16 and CPU cores)
redis-1  | 1:M 12 Oct 2026 11:55:29.988 * <search> Redis version found by RedisSearch : 8.10.
2 - oss
redis-1  | 1:M 12 Oct 2026 11:55:29.988 * <search> RediSearch version 8.10.1 (Git=)
redis-1  | 1:M 12 Oct 2026 11:55:29.988 * <search> gc: ON, prefix min length: 2, min word len
gth to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return,
oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, max doctable size:
1000000, max number of search results:  1000000, default scorer: BM25STD,
redis-1  | 1:M 12 Oct 2026 11:55:29.988 * <search> Initialized thread pools!
redis-1  | 1:M 12 Oct 2026 11:55:29.988 * <search> Enabled workers threadpool of size 8
redis-1  | 1:M 12 Oct 2026 11:55:29.989 * <search> Subscribe to config changes
redis-1  | 1:M 12 Oct 2026 11:55:29.989 * <search> Subscribe to cluster slot migration events
redis-1  | 1:M 12 Oct 2026 11:55:29.989 * <search> Subscribe to cluster topology change event
s
redis-1  | 1:M 12 Oct 2026 11:55:29.989 * <search> Enabled role change notification
redis-1  | 1:M 12 Oct 2026 11:55:29.989 * <search> Cluster configuration: AUTO partitions, ty
pe: 0, coordinator timeout: 0ms
redis-1  | 1:M 12 Oct 2026 11:55:29.989 * Module 'search' loaded from /usr/local/lib/redis/mo
dules//redisearch.so
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries> RedisTimeSeries version 81001, git_sha
=
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries> Redis version found by RedisTimeSeries
 : 8.10.2 - oss
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries> Registering configuration options: [
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries>  { ts-compaction-policy   :
   }
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries>  { ts-num-threads         :
 3 }
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries>  { ts-libmr-protocol      :     INTERN
AL }
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries>  { ts-retention-policy    :
 0 }
redis-1  | 1:M 12 Oct 2026 11:55:29.990 * <timeseries>  { ts-duplicate-policy    :        blo
ck }
```

Multi-container app started — Ran docker-compose up; logs showed both the web and redis containers building/starting and joining the same Compose-managed network.

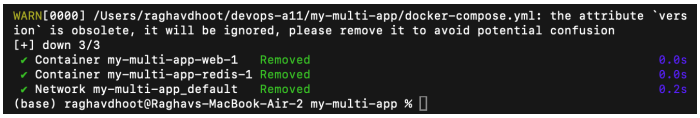


Hello, Docker! This page has been viewed 7 times.



Hello, Docker! This page has been viewed 14 times.

Cross-service communication verified — Accessed http://localhost:8080 and refreshed repeatedly; the hit counter incremented on each request, confirming the web container successfully reached redis over the internal Docker network using the service name as hostname.

	 Teardown — Ran docker-compose down to stop and remove the containers and network (volume retained by default).